TITLE PAGE

- **Problem Statement ID –01**
- **Problem Statement Title- ISOTHERM**- Eco-Loop Building Agents
- **Theme-** AI-Powered Autonomous Smart Building Optimization Challenge
- **PS Category-** Software
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IDEA TITLE

Idea Title: ISOTHERM — Decoupled Stdio MCP Bus for Closed-Loop Physical AI Building Control

Proposed Solution Architecture:

- Decoupled AI Control: Couples EnergyPlus 24.1.0 physics engine with local Llama 3.1 8B via Model Context Protocol (MCP) over standard I/O pipes.
- 3-Hour Synchronous Callback: Blocks physics execution every 3 simulated hours, eliminating stale-setpoint drift and reducing LLM inference cycles by 87.5%.
- Canonical Operating Matrix: Evaluates a 7-quadrant decision table (Season × Occupancy × Utility Tariff) with real-time comfort self-correction.

Innovation & Uniqueness:

- Joule-Exact Meter Reconciliation: Solves pyenergyplus warmup contamination to achieve 0.0000 J discrepancy against official compilation meters across 960 rows.
- The "Thermal Battery" Discovery: Uses morning mid-peak gas ($0.08/kWh) to warm building concrete mass, coasting during afternoon on-peak ($0.15/kWh) for a +2.9 pp comfort win while shedding load.

TECHNICAL APPROACH

Core Technologies & Engineering Stack:

- Physics Engine: EnergyPlus 24.1.0 (pyenergyplus C++ runtime API with custom callback hooks).
- AI Orchestration: Local Llama 3.1 8B-Instruct (Ollama) executing tools over zero-latency Stdio MCP transport.
- State Bus: SQLite 3 (sim_state.db) capturing 15-min telemetry, atomic action queues, and decision audit logs.
- Presentation Dashboard: Standalone React 18 + Vite minimal luxury web app reading frozen database records with zero runtime overhead.

Methodology & 4-Layer Defense-in-Depth Safety Hierarchy:

- Layer 1 (Schema Validation): Pydantic JSON parser intercepts syntax errors and malformed tool calls before execution.
- Layer 2 (Timeout Fallback): 60-second sub-process timer injects canonical setback defaults (16° C/27° C) if GPU inference throttles.
- Layer 3 (Hard Code Clamps): Deterministic Python bounds enforce 16–24° C heating, 22–30° C cooling, and >= 2° C deadband before API writes.
- Layer 4 (Spatial Independence): 10 independent schedule compact handles eliminate global schedule conflicts across all 5 zones.

FEASIBILITY AND VIABILITY

Feasibility & Computational Efficiency:

- 100% Local Enterprise Execution: Operates on standard hybrid CPU/GPU hardware without cloud API dependencies or recurring token costs.
- Low Runtime Overhead: 3-hour control cadence completes multi-day building simulations in under 2 minutes of wall-clock time.

Potential Challenges & Engineered Mitigations:

- Challenge: LLM Hallucinations — Free-form prompts risk out-of-bounds setpoints or rapid valve cycling that crashes C++ solver convergence.

  -> Mitigation: Restricted LLM role to canonical decision-table evaluation with comfort-feedback self-correction overrides.

- Challenge: Telemetry Contamination — Silent EnergyPlus warmup days pollute database logging and distort baseline savings.

  -> Mitigation: Implemented explicit pyenergyplus warmup interlock (warmup_flag guard), achieving Joule-exact meter reconciliation.

- Challenge: VAV Reheat Waste — Perimeter cooling over-cools interior core zones, forcing hot-water reheat coils to fight central chillers.

  -> Mitigation: Enforced season-aware prompt clamping (16.0° C summer floor), achieving 100% elimination of summer reheat waste (0.00 kW peak reheat).

ARTIFACTS

Verified Ground-Truth Quantitative Scorecard (Jan 15 & Jul 1 Representative Days):

- Joule-Exact Meter Reconciliation: 0.0000 J discrepancy between SQLite database sums and official EnergyPlus eplusmtr.csv compiled meters across 960 rows.
- 100% Summer Reheat Elimination: Reheat gas consumption reduced from 300+ kWh down to literally 0.00 kWh (0.00 kW peak reheat demand).
- Summer Occupied Comfort: 34.5% compliance within ASHRAE 55 comfort band (PMV -0.5 to +0.5), protecting tenant comfort while saving 100% of reheat gas.
- Winter Occupied Comfort: 14.5% AI Control vs. 12.3% Baseline (+2.3 percentage point overall comfort win against sub-zero Chicago weather).
- Winter Peak Comfort Win: Achieved a +2.9 percentage point comfort win during afternoon On-Peak (12:00–18:00) at $0.15/kWh by coasting on morning thermal battery storage.

Challenge Deliverables Included in Submission:

- Code & Models: Full Python codebase (src/, scripts/) and configured EnergyPlus .idf building models (building_model/).
- Dashboard & Documentation: Standalone React/Vite luxury UI (ISOTHERM dashboard), System Architecture whitepaper, and 2m23s video demonstration.

RESEARCH AND REFERENCES

Industry Standards & Engineering Guidelines:

- ASHRAE Standard 55-2020: Thermal Environmental Conditions for Human Occupancy (Defines Fanger PMV comfort boundary [-0.5, +0.5] and seasonal clo parameters).
- ASHRAE Guideline 36-2018: High-Performance Sequences of Operation for HVAC Systems (Principles of VAV box reheat elimination, deadband enforcement, and optimal start).

Technical Frameworks & Protocols:

- Model Context Protocol (MCP) Specification: Stdio transport architecture for sandboxed process isolation and zero-overhead inter-process communication with LLM engines.
- EnergyPlus 24.1.0 Engineering Reference: Runtime API (pyenergyplus) callback hooks, zone air balance thermodynamics, and Schedule:Compact actuator manipulation.
- Chicago Time-of-Use (TOU) Commercial Tariffs: Electric rate structures ($0.05/kWh off-peak, $0.08/kWh mid-peak, $0.15/kWh on-peak) used for cost optimization.